

ADMET Prediction and Molecular Docking Analysis Report

Overview

This report evaluates the pharmacokinetic and toxicological profiles of four key phytochemicals: Berberine, Quercetin, Curcumin, and Resveratrol. The analysis was conducted using pkCSM, covering physicochemical properties, absorption, distribution, metabolism, excretion, and toxicity (ADMET). Additionally, molecular docking results against Escherichia coli DNA gyrase B (PDB ID: 5L3J) are provided.

1. Berberine

1.1 Physicochemical Properties

- Molecular Weight: 336.37 Da
- LogP: 3.10
- H-Bond Acceptors: 4
- H-Bond Donors: 0
- Rotatable Bonds: 2

1.2 Absorption

- Water Solubility: $-3.973 \log \text{ mol/L}$
- Caco-2 Permeability: 1.734
- Human Intestinal Absorption: 97.147%
- P-glycoprotein Substrate: Yes

Berberine exhibits excellent intestinal absorption and high membrane permeability, suggesting favorable oral bioavailability.

1.3 Metabolism & Excretion

Berberine may participate in several CYP-mediated drug interactions due to its inhibitory activity toward CYP1A2, CYP2D6, and CYP3A4. The high predicted clearance indicates efficient elimination from the systemic circulation.

1.4 Toxicity

Despite its strong antibacterial binding, the predicted mutagenicity and hepatotoxicity necessitate further safety evaluation.

1.5 Molecular Docking Performance

Binding Affinity: -8.034 kcal/mol. Berberine produced the strongest binding affinity, indicating superior interaction with the DNA gyrase B active site.

2. Quercetin

2.1 Physicochemical Properties

- Molecular Weight: 302.24 Da
- LogP: 1.99
- H-Bond Acceptors: 7
- H-Bond Donors: 5
- Rotatable Bonds: 1

2.2 Absorption

- Water Solubility: -2.925 log mol/L
- Caco-2 Permeability: -0.229
- Human Intestinal Absorption: 77.207%
- P-gp Substrate: Yes

Quercetin shows acceptable oral absorption with moderate membrane permeability.

2.3 Metabolism & Excretion

Quercetin presents a lower risk of drug-drug interactions compared to Berberine. Predicted clearance is within the normal range.

2.4 Toxicity

Quercetin demonstrates a highly favorable safety profile.

2.5 Molecular Docking Performance

Binding Affinity: -7.635 kcal/mol. Quercetin shows strong binding to DNA gyrase B while maintaining low toxicity.

3. Curcumin

3.1 Physicochemical Properties

- Molecular Weight: 368.39 Da
- LogP: 3.37
- H-Bond Acceptors: 6

- H-Bond Donors: 2
- Rotatable Bonds: 8

3.2 Absorption

- Water Solubility: $-4.01 \log \text{ mol/L}$
- Caco-2 Permeability: -0.093
- Human Intestinal Absorption: 82.19%
- P-gp Substrate: Yes

Curcumin demonstrates good intestinal absorption but limited aqueous solubility.

3.3 Metabolism & Excretion

Potential metabolic interactions are likely due to inhibition of multiple CYP enzymes. Relatively slow excretion is predicted.

3.4 Toxicity

Curcumin shows a favorable predicted toxicity profile.

3.5 Molecular Docking Performance

Binding Affinity: -6.816 kcal/mol . Curcumin displayed moderate binding affinity.

4. Resveratrol

4.1 Physicochemical Properties

- Molecular Weight: 228.25 Da
- LogP: 2.97
- H-Bond Acceptors: 3
- H-Bond Donors: 3
- Rotatable Bonds: 2

4.2 Absorption

- Water Solubility: $-3.178 \log \text{ mol/L}$
- Caco-2 Permeability: 1.17
- Human Intestinal Absorption: 90.935%
- P-gp Substrate: Yes

Resveratrol demonstrates high intestinal absorption and favorable membrane permeability.

4.3 Metabolism & Excretion

May inhibit CYP1A2 and CYP2C19 enzymes. Steady elimination from the body.

4.4 Toxicity

While non-hepatotoxic, Resveratrol may possess mutagenic potential according to AMES predictions.

4.5 Molecular Docking Performance

Binding Affinity: -6.763 kcal/mol. Resveratrol exhibited the weakest binding affinity among the compounds tested.

Overall Prediction

Among the four phytochemicals, **Berberine** demonstrated the most potent predicted binding affinity toward **E. coli DNA gyrase B (-8.034 kcal/mol)** and superior oral absorption. However, potential mutagenicity and hepatotoxicity indicate a need for rigorous safety testing. **Quercetin** emerged as the most balanced candidate, offering a strong binding affinity (-7.635 kcal/mol) alongside an excellent predicted safety profile and low risk of drug interactions. In conclusion, while **Berberine** is the most potent inhibitor based on docking scores, **Quercetin** provides the most promising efficacy-to-safety ratio for further experimental validation.

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